

DS & AI ENGINEERING

Algorithms

Graphs Algorithms

Lecture No.- 01



By- Aditya sir

Recap of Previous Lecture



Topic

Topic

Dynamic Programming



Topics to be Covered



Topic

Topic

Topic

Graph Algorithms

↳ Traversals



About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.



Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW

TTTe

Update:-

Sunday → 27th July (No Algo class)

Graph Traversal



Topic : Graphs – Traversals:

Graph Traversals:

Graph Traversal is a process of visiting each and every node of the tree/graph in some specific order and processing the information exactly once.



Topic : Graphs – Traversals:

Graph Traversals Content:

(1) Depth First Search (DFS) / Depth First Traversal (DFT)

A. Undirected graph

- a. Connected graph
- b. Disconnected graph.

B. Directed graph

- a. Directed Acyclic Graph (DAG)
- b. Topological Sorting / *ordering*



Topic : Graphs – Traversals:

Graph Traversals Content:

(1) Breadth First Search / Breadth First Traversals
(BFS) (BFT)

a. FIFO BFS (default / standard Approach)

b. LIFO BFS (bonus)

Stack → LIFO → DFS

Queue → FIFO → BFS



Topic : Graphs – Traversals:

Applications of DFS ~~and BFS~~:

1. Topological Sorting /ordering
2. Connected Components (cc)
3. Strongly Connected Components (SCC)
4. Bi-Connected Components (BCC)
5. Articulation Point / Cut Vertex
6. Properties

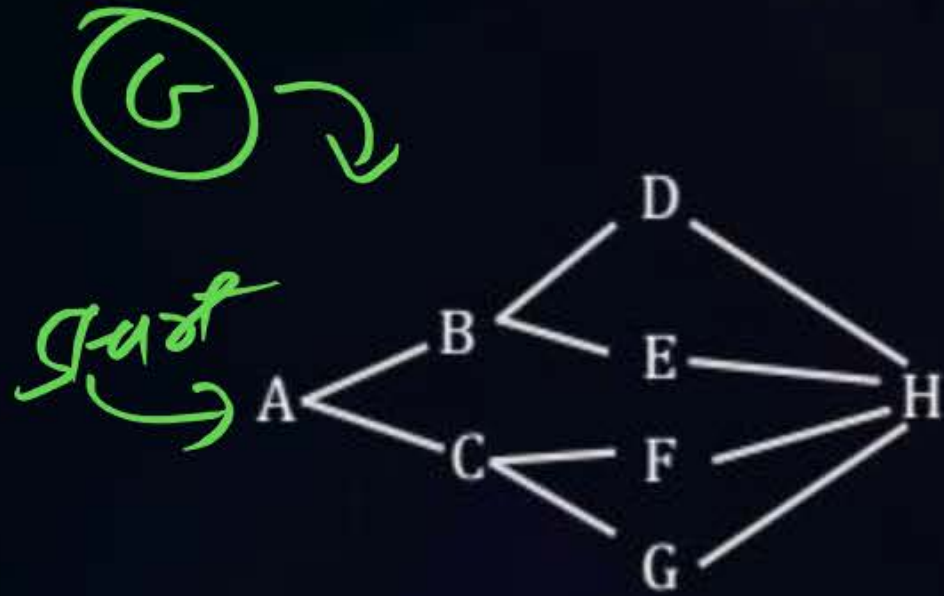


Topic : Graphs – Traversals:

Breadth First Traversals (BFT) :

(a) FIFO BFS/BFT (default → Standard)

Given a graph: $G(V, E)$



BFS Spanning Tree

BFS Starting at A:



O/P:



Node
Parent

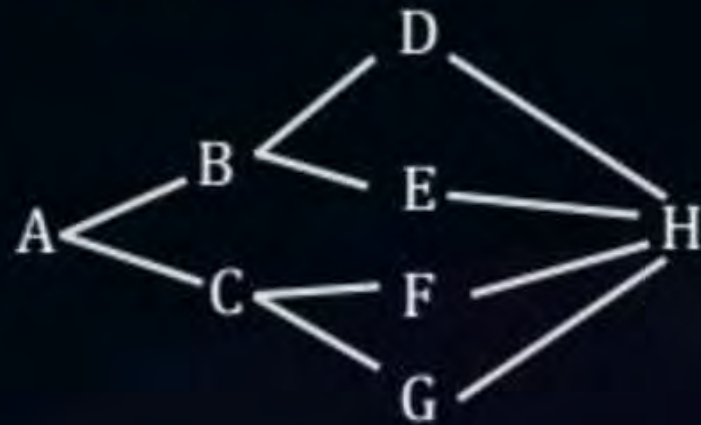
B	C	D	E	G	F	H
A	A	B	B	<u>C</u>	C	<u>D</u>

FIFO Queue



Topic : Graphs – Traversals:

Observation:



A → D:

1. A — B — D; 1 + 1 = 2

2. A — B — E — H — D : 4

and many more

Property:

In a uni-weighted graph (every edge has equal weight) or in an unweighted graph, BFS solves the problem of SSSP Single Source Shortest Path from starting vertex to all other vertex.



Topic : Graphs – Traversals:

Bonus:

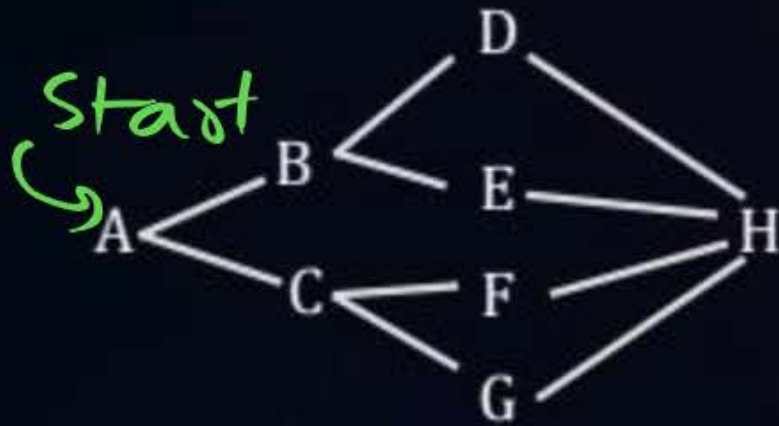
LIFO BFS

(Special Case)

different from DFS

Here ~~are~~^{we} will follow the BFS strategy but in LIFO order.

LIFO BFS Spanning Tree Starting at A.



O/P:

A C G H E D F B

Node
Parent

B	C	F	G	H	D	E
A	A	C	C	G	H	H

LIFO
~~FIFO~~ Queue



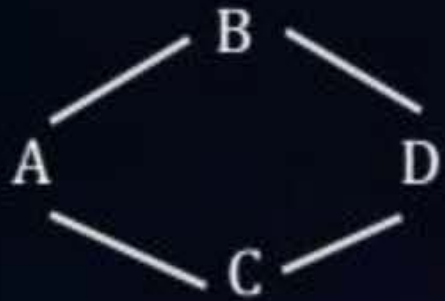
Topic : Graphs – Traversals:

Depth First Traversal (DFT)

A — B — C — D

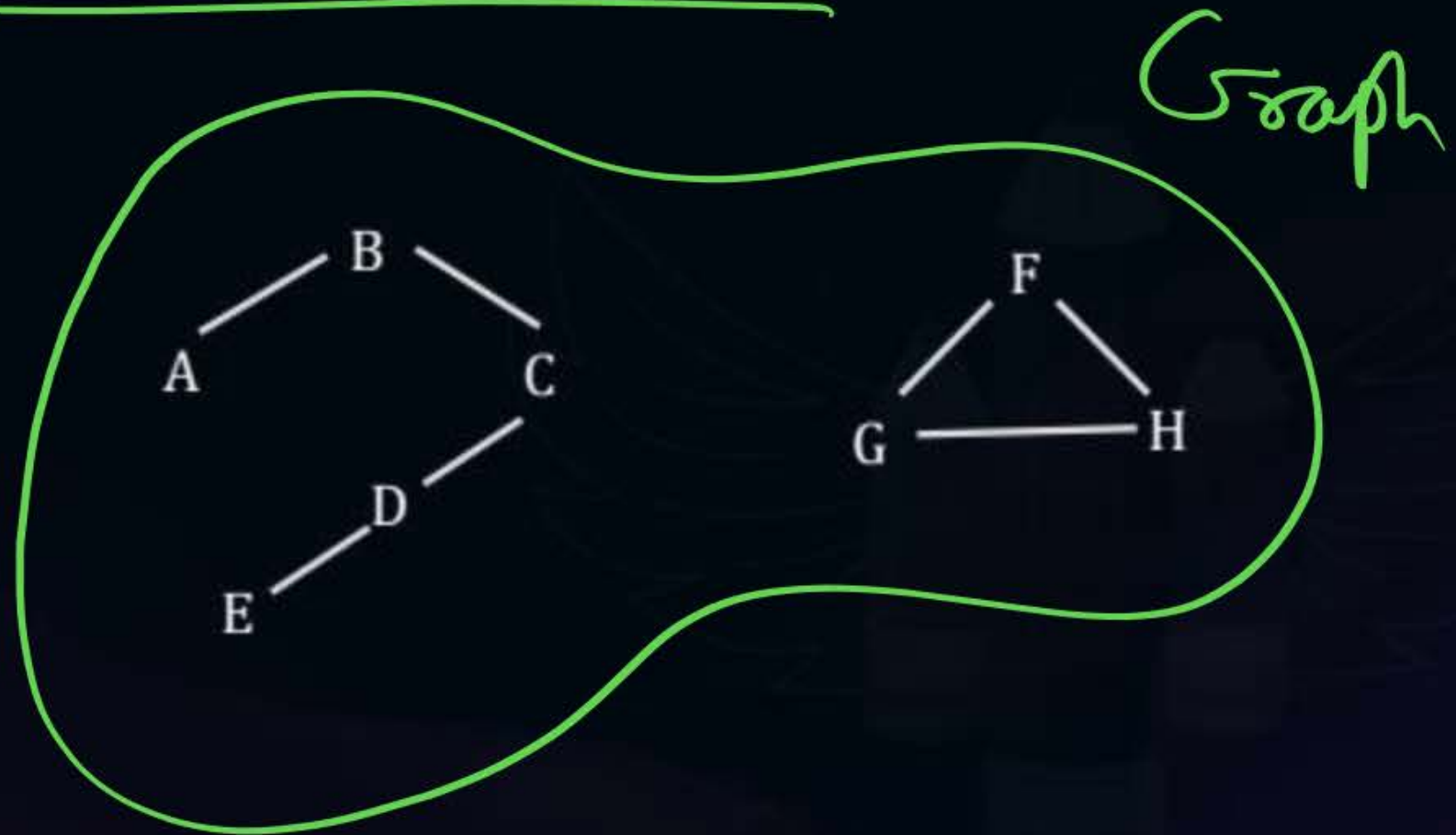
(1) Connected graph

$G(V, E)$



1 CC

(2) Disconnected graph





Topic : Graphs – Traversals:

Depth First Traversal (DFT) on Undirected Graph a. Connected.

Terminologies:

1. E-node: (Exploring Node):

→ Node which is currently getting explored.

2. Live Node:

→ Node that is not yet fully explored.

(Live nodes are present/kept in stack.)

3. Dead Node:

→ Node that has already been fully explored (already popped out from stack).



Topic : Graphs – Traversals:



$V \cdot V \cdot V \cdot \text{Temp}$

During Traversal, the different types of timing values that a node is associated with:

1. Discovery Time: $d(x)$:

→ It is the time at which the node has got discovered / visited for the very first time.

2. Finishing Time: $f(x)$

→ It is the time at which the node becomes a dead node.



Topic : Graphs – Traversals:

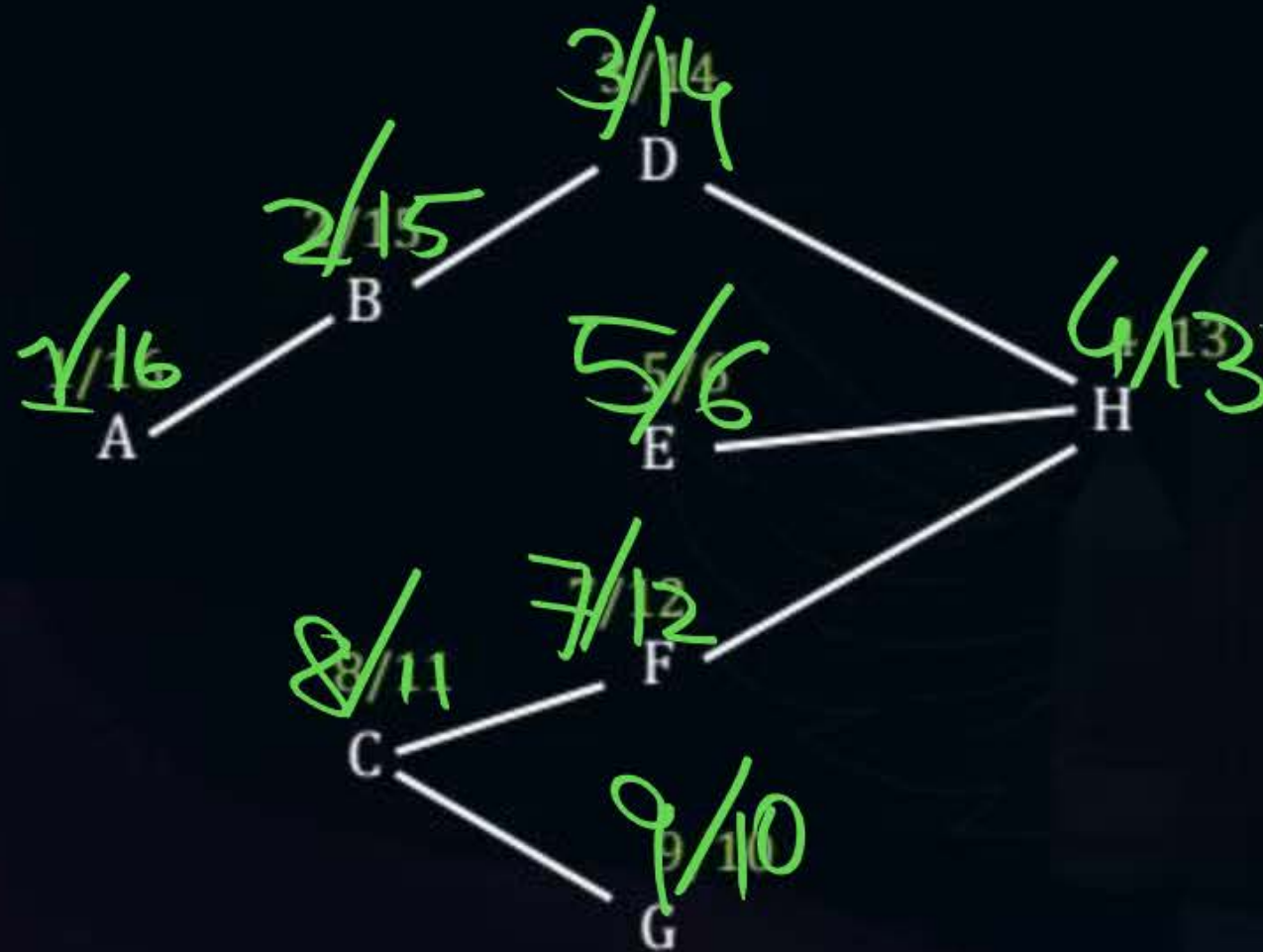
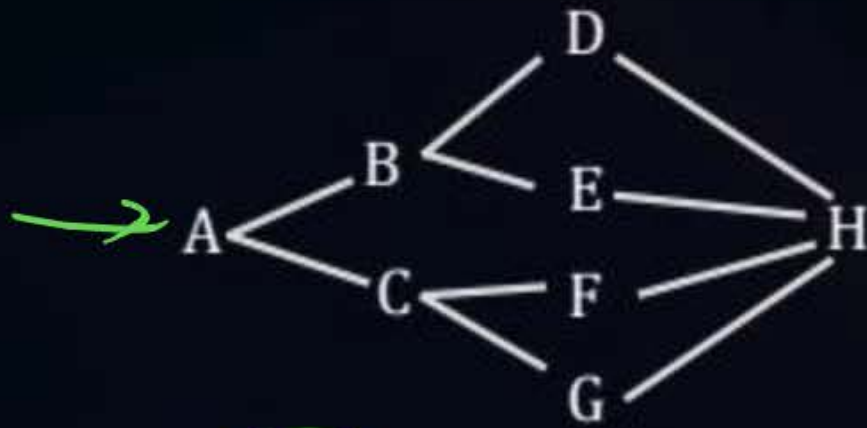


d/f

A

E.g.1 DFS in an undirected connected graph.

DFS Spanning Tree starting at A



$$8 \times 2 = \underline{\underline{16}}$$

A	G
E	F
H	
D	
B	
A	

Stack

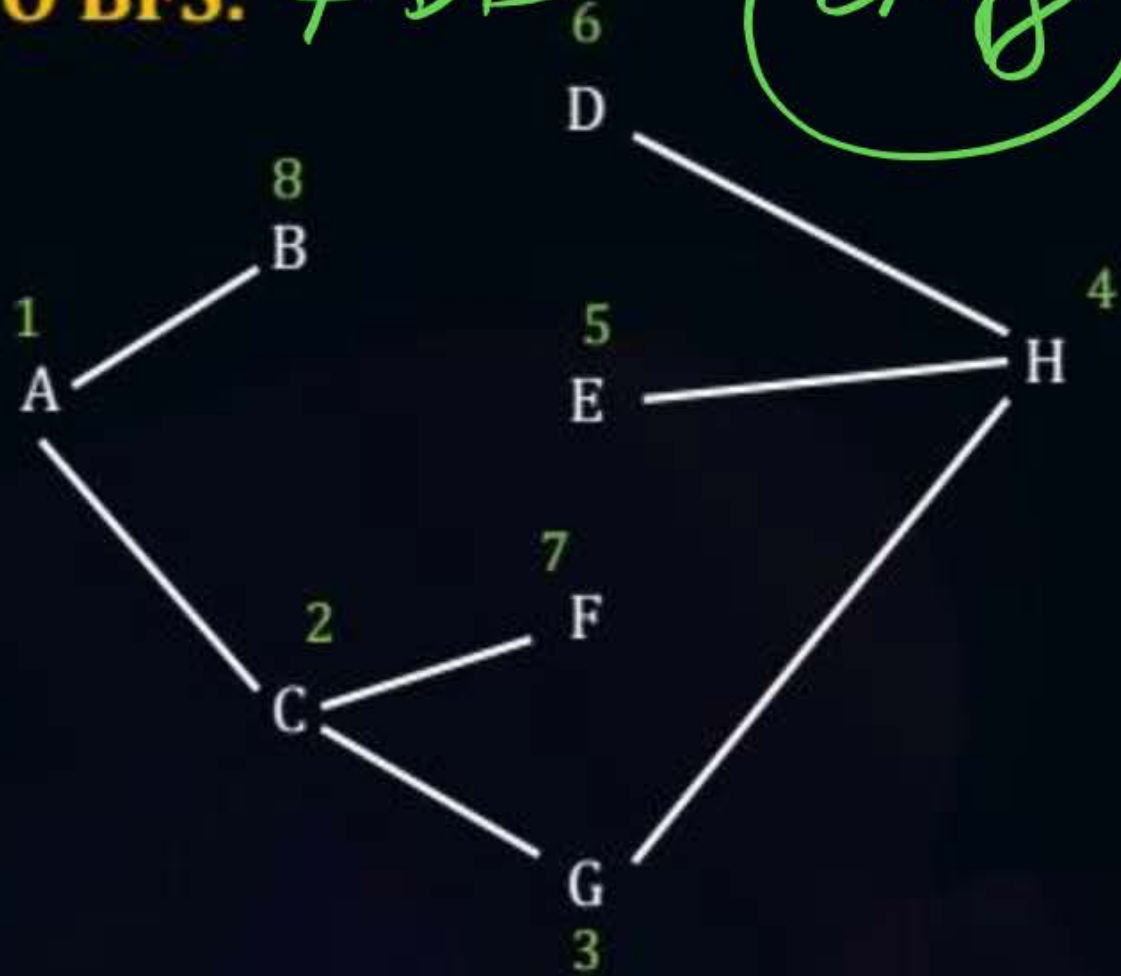
O/P: A B D H E F C G



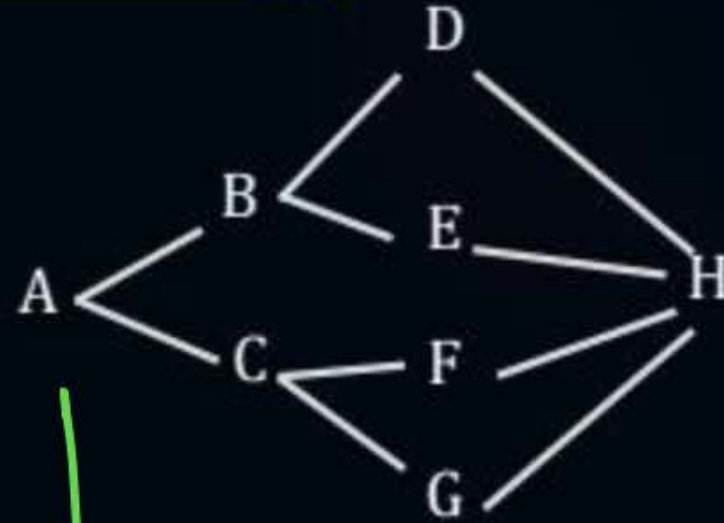
Topic : Graphs – Traversals:

LIFO BFS: $\neq$ DFS

diff



O/P: A B D H E F C G



DFS O/P: A C G H E B D F

[MCQ]



#Q. Which of the following represent valid DFS ~~DFS~~ traversal on graph G?

② DFS?

A

ABFEDGHC

DFS
X

X

B

ABEHDFCG

DFS
✓

X

C

DHGCFABE

DFS
✓

X

D

HEBDACGF

DFS
✓

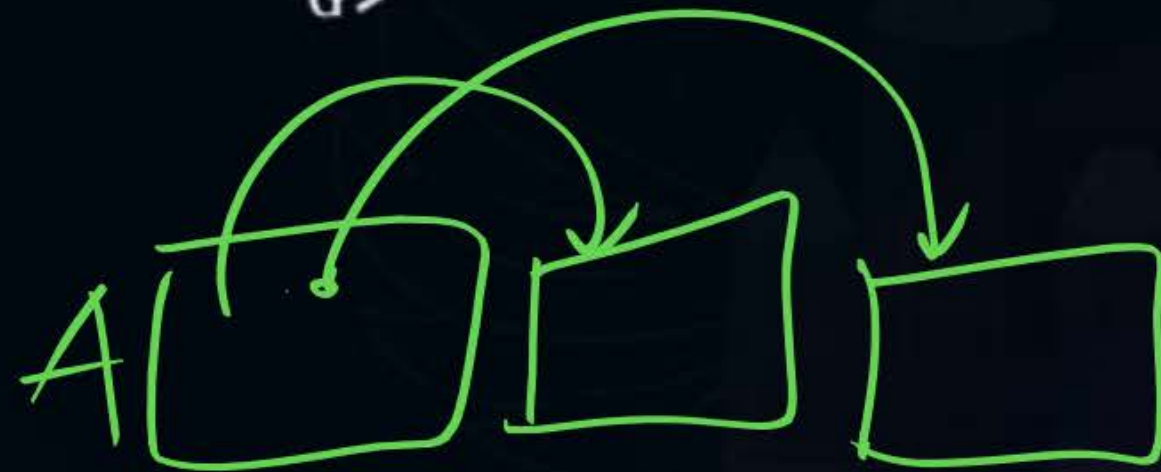
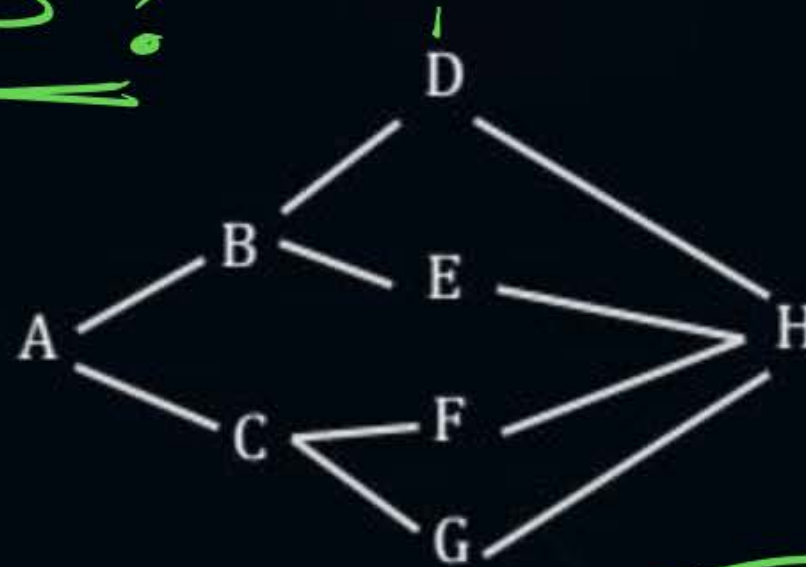
X

E

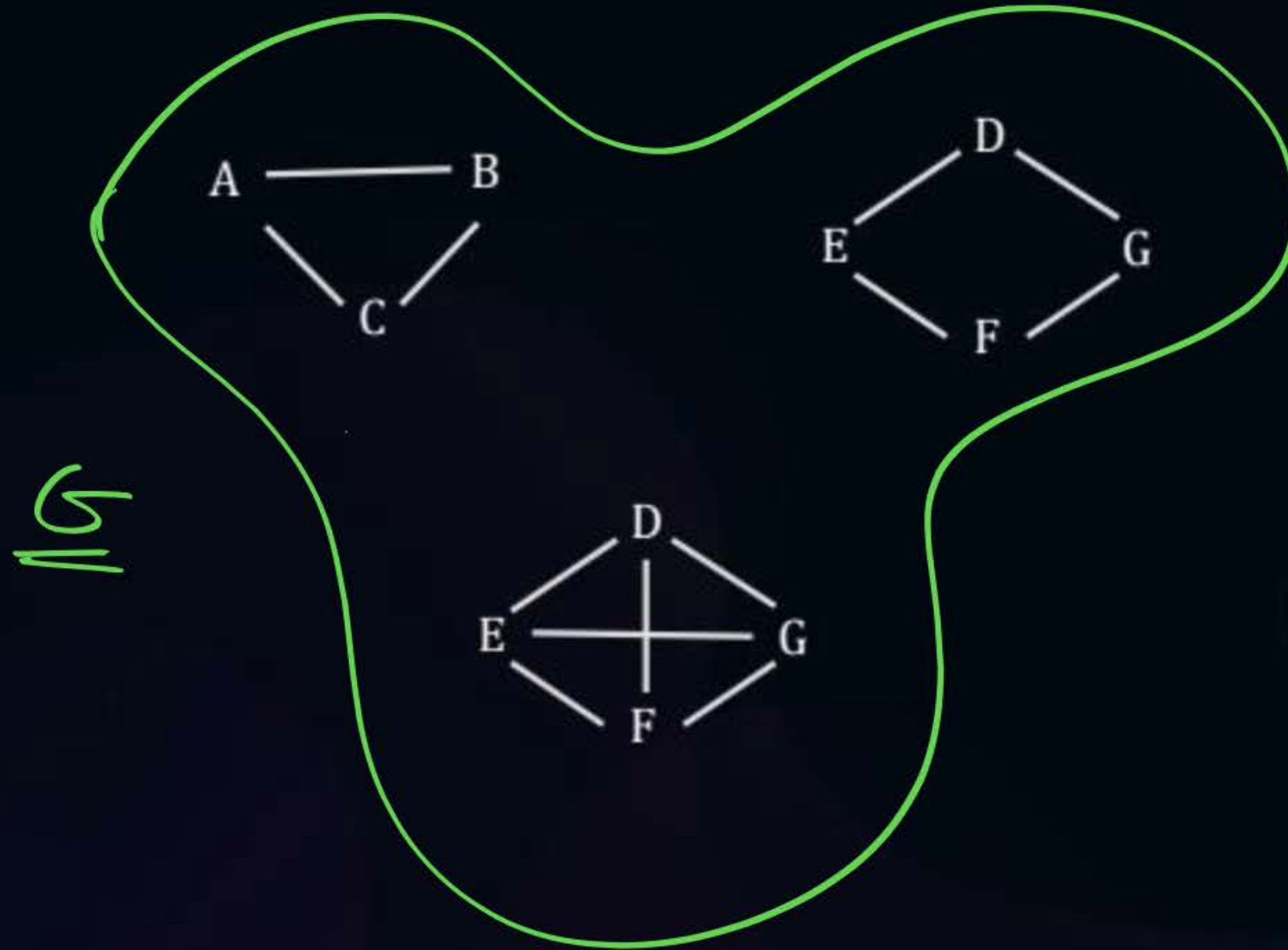
HDBEFCGA

DFS
X

X



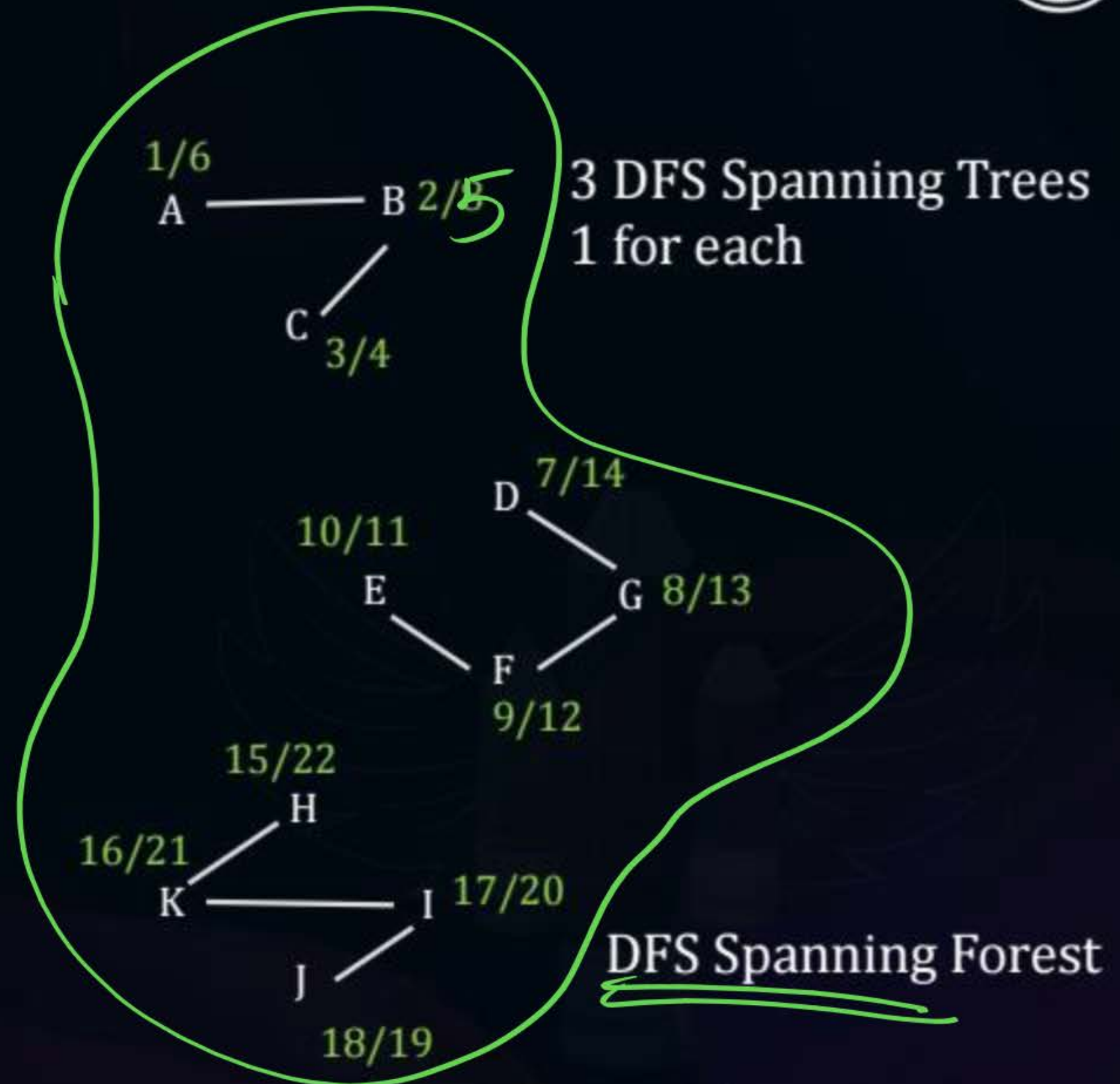
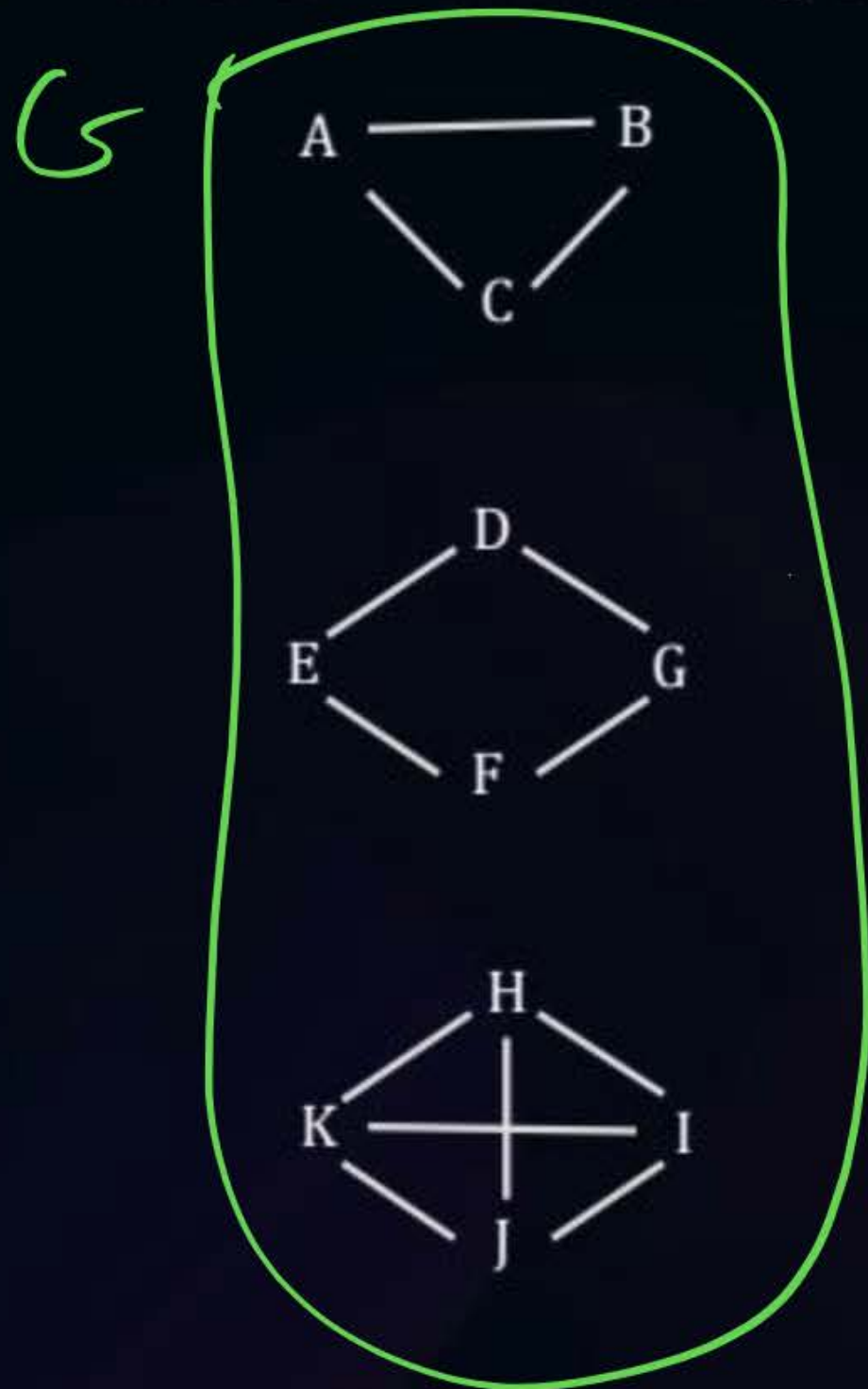
#Q. DFS / DFT on a undirected Disconnected graph:



Given $G(V, E)$

Disconnected

Sol. G: 3 connected components



Type of concept in GATE:

Application of Discovery and Finishing Time)

#Q. You are given a graph G with 5 vertices $\{A, B, C, D, E\}$

The Discovery and Finishing time (d, f) for each node/vertex after carrying out DFS are given as follows:

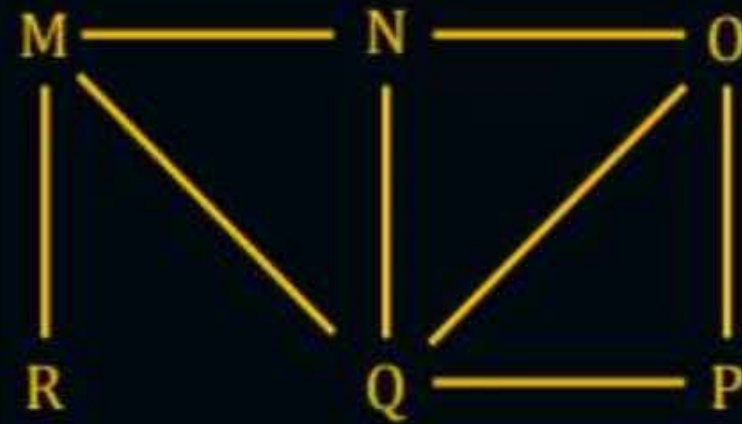
1. Is G connected / disconnected? (for each option)
2. How many connected components in G ?
3. What are the vertices in each of those connected components?

Sol. (d, ~~f~~)

	A	B	C	D	E
✓ 1.	(1, 2)	(3, 5)	(7, 8)	(11, 12)	(18, 20)
✓ 2.	(1, 15)	(3, 13)	(4, 12)	(5, 11)	(6, 9)
✓ 3.	(1, 13)	(2, 7)	(16, 18)	(19, 25)	(21, 23)
4.	(1, 16)	(3, 11)	(5, 10)	(20, 25)	(22, 23)

1. Disconnected {A} {B} {C} {D} {E} → 5 components
2. Connected graph {A, B, C, D, E} → 1 component)
3. Disconnected graph {A, B} {C} {D, E} → 3 components
4. Disconnected graph {A, B, C} {D, E} → 2 components

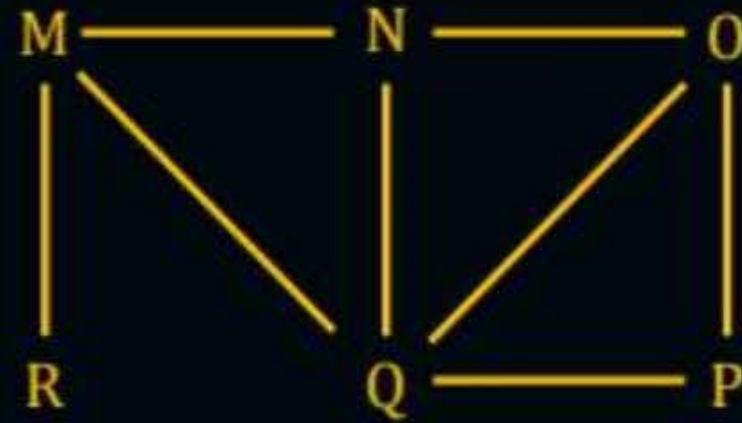
#Q. Which are valid DFS traversals?



- A** M N O P Q R
- B** N Q M P O R
- C** Q M N R O P
- D** P O Q N M R

#Q. Which are valid DFS traversals?

38.46%



A M N O P Q R

B N Q M P O R

C Q M N R O P

D P O Q N M R

E) NONE

DFS

BFS

✓

✗

✗

✗

✗

✗ m

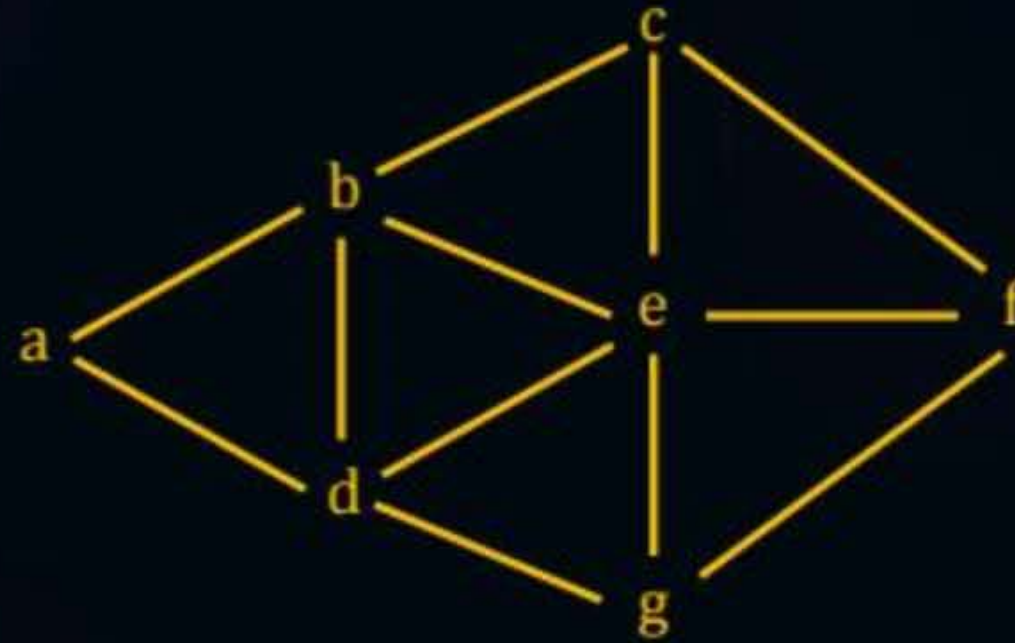
✓

✓

[P O Q N M R]

~~P~~ ~~O~~ ~~Q~~ ~~N~~ ~~M~~ ~~R~~

#Q. Given



Valid DFS ?

A a b e f d g e

B a b e f c g d

C a d g e b c f

D a d b c g e f

HW



Summary



BFS

DFS



THANK - YOU